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Experiment 1:

```
clc;
clear;

% Experiment 1: FDMA, TDMA and CDMA

% System Parameters
totalBandwidthHz = 1e6;
fdmaChannelHz = 200e3;
tdmaSlots = 8;
processingGain = 64;
targetSIRdB = 7;

% Capacity Estimation

fdmaUsers = floor(totalBandwidthHz/fdmaChannelHz);
tdmaUsers = tdmaSlots;
targetSIR = 10^(targetSIRdB/10);
cdmaUsers = floor(1 + processingGain/targetSIR);

fprintf('Maximum Supported Users\n');
fprintf('FDMA: %d users\n',fdmaUsers);
fprintf('TDMA: %d users\n',tdmaUsers);
fprintf('CDMA: %d users\n',cdmaUsers);

method = {'FDMA';'TDMA';'CDMA'};
supportedUsers = [fdmaUsers;tdmaUsers;cdmaUsers];
```

```
results = table(method,supportedUsers);
disp(results);

% CDMA SIR Analysis
K = 2:30;
sirLinear = processingGain./(K-1);
sirDb = 10*log10(sirLinear);

figure;
plot(K,sirDb,'bo-','LineWidth',1.5);
hold on;
plot(K,targetSIRdB*ones(size(K)),'r--');
xlabel('Number of Simultaneous Users');
ylabel('SIR (dB)');
title('CDMA SIR versus Number of Users');
legend('Estimated SIR','Target SIR');
grid on;
hold off;

% Capacity Comparison
figure;
bar(supportedUsers);
set(gca,'XTickLabel',method);
ylabel('Number of Users');
title('FDMA, TDMA and CDMA Capacity');
grid on;

% Mobile Generation Comparison
generation = {'1G';'2G';'2G';'3G';'4G';'5G'};
system = {'AMPS';'GSM';'IS-95';'UMTS';'LTE';'5G NR'};
accessMethod = {'FDMA';'TDMA';'CDMA';'WCDMA';'OFDMA';'OFDMA-based'};
```

```
keyIdea = {
    'Separate frequency channels'
    'Separate time slots'
    'Code-based user separation'
    'Wideband code-based access'
    'Orthogonal frequency resources'
    'Flexible OFDM-based radio access'
    };

evolutionTable = table(generation,system,accessMethod,keyIdea);
disp(evolutionTable);

% Resource Allocation Diagrams
figure;
colors = lines(4);

% FDMA
subplot(1,3,1);
hold on;
for u = 1:4
    rectangle('Position',[0,u-1,1,0.8], ...
        'FaceColor',colors(u,:),'EdgeColor','k');
    text(0.5,u-0.6,sprintf('User %d',u), ...
        'HorizontalAlignment','center');
end
xlim([0 1]);
ylim([0 4]);
xlabel('Time');
ylabel('Frequency');
title('FDMA');
set(gca,'YTick',[]);
```

```
hold off;

% TDMA
subplot(1,3,2);
hold on;
for u = 1:4
    rectangle('Position',[u-1,0,0.8,1], ...
        'FaceColor',colors(u,:),'EdgeColor','k');
    text(u-0.6,0.5,sprintf('User %d',u), ...
        'HorizontalAlignment','center');
end
xlim([0 4]);
ylim([0 1]);
xlabel('Time Slots');
ylabel('Frequency');
title('TDMA');
set(gca,'YTick',[]);
hold off;

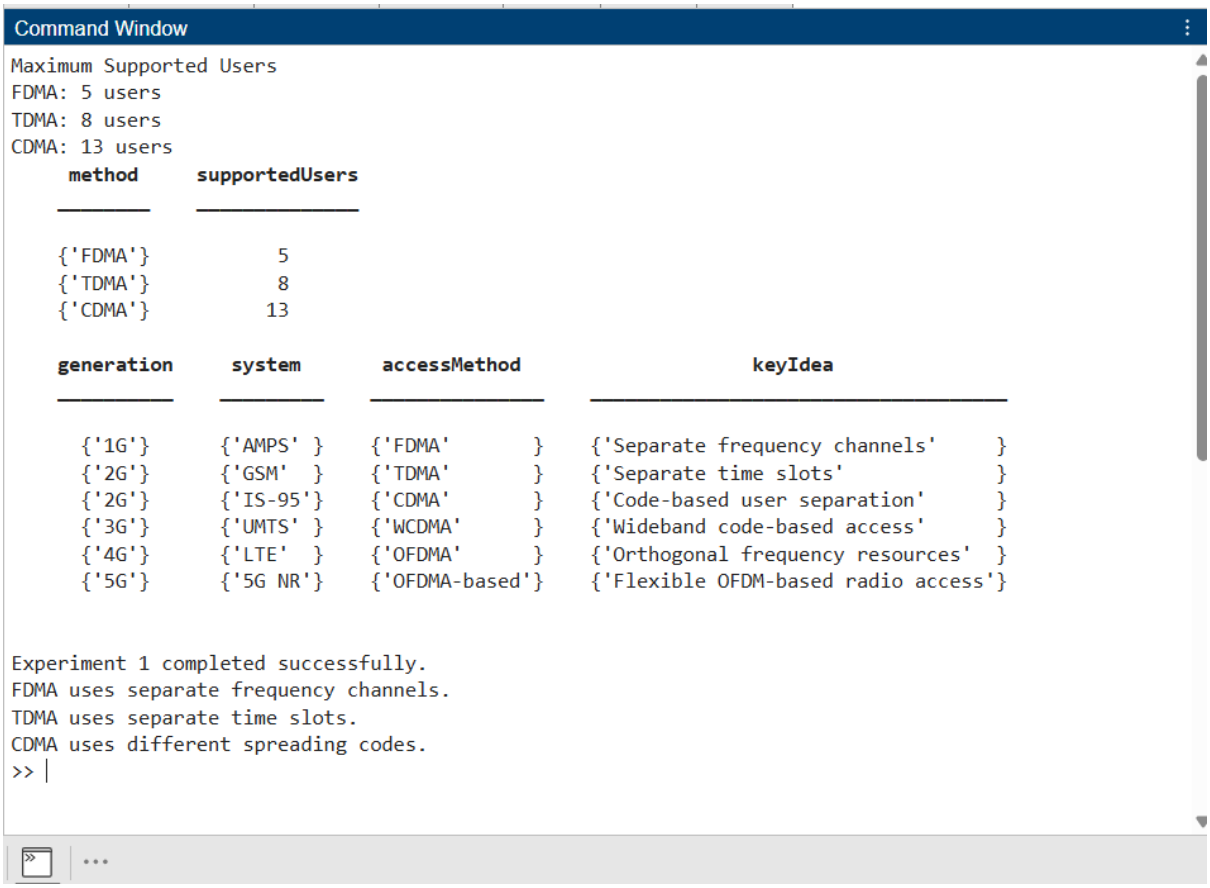
% CDMA
subplot(1,3,3);
hold on;
for u = 1:4
    rectangle('Position',[0,0,1,1], ...
        'FaceColor',colors(u,:), ...
        'FaceAlpha',0.2,'EdgeColor',colors(u,:));
end
text(0.5,0.6,'Users 1-4 share', ...
    'HorizontalAlignment','center');
text(0.5,0.4,'band using codes', ...
    'HorizontalAlignment','center');
xlim([0 1]);
```

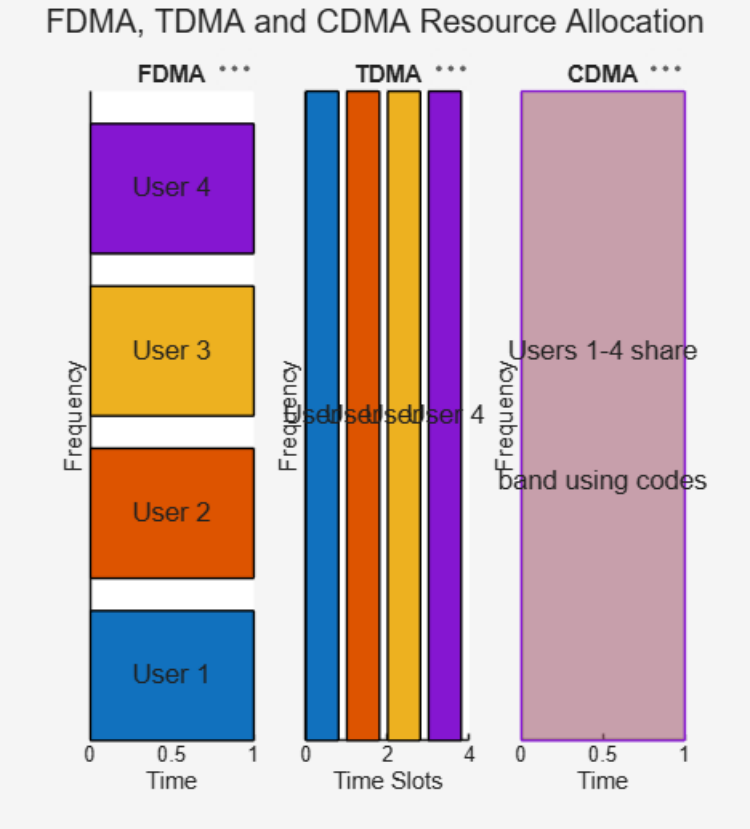
```
ylim([0 1]);
xlabel('Time');
ylabel('Frequency');
title('CDMA');
set(gca,'YTick',[]);
hold off;

sgtitle('FDMA, TDMA and CDMA Resource Allocation');

% Conclusion
fprintf('\nExperiment 1 completed successfully.\n');
fprintf('FDMA uses separate frequency channels.\n');
fprintf('TDMA uses separate time slots.\n');
fprintf('CDMA uses different spreading codes.\n');
```

Implementation:





Experiment 2:

```
clear
clc

serviceAreaKm2 = 5;
radiiKm = [1.0 0.5];
usersPerCell = 30;

hexArea = (3*sqrt(3)/2)*radiiKm.^2;
cellCount = ceil(serviceAreaKm2./hexArea);
totalUsers = cellCount*usersPerCell;

Result = table(radiiKm',hexArea',cellCount', ...
    totalUsers','VariableNames', ...
    {'Radius_km','CellArea_km2','NumberOfCells','TotalUsers'});
disp(Result)
```

```
R = 1;
centers = [0 0;
    1.5*R sqrt(3)*R/2;
    1.5*R -sqrt(3)*R/2;
    0 sqrt(3)*R;
    0 -sqrt(3)*R;
    -1.5*R sqrt(3)*R/2;
    -1.5*R -sqrt(3)*R/2];

theta = (0:6)*pi/3;

figure
hold on
axis equal
grid on

for k = 1:size(centers,1)
    x = centers(k,1)+R*cos(theta);
    y = centers(k,2)+R*sin(theta);
    plot(x,y,'k')
    text(centers(k,1),centers(k,2), ...
        sprintf('Cell %d',k), ...
        'HorizontalAlignment','center')
end

xlabel('X (km)')
ylabel('Y (km)')
title('Seven-Cell Hexagonal Network')
hold off

mobileX = linspace(-0.2*R,1.7*R,300);
mobileY = zeros(size(mobileX));
```

```
d1 = hypot(mobileX,mobileY);
d2 = hypot(mobileX-1.5*R,mobileY-sqrt(3)*R/2);

servingCell = ones(size(mobileX));
servingCell(d2 < d1) = 2;

handoffIndex = find(diff(servingCell)~=0,1);
handoffX = mobileX(handoffIndex);

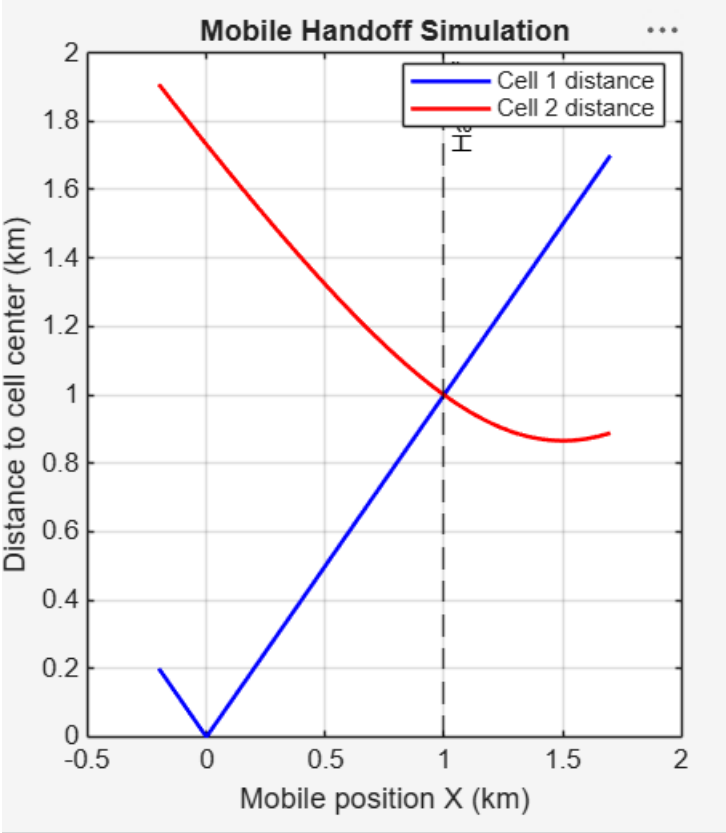
figure
plot(mobileX,d1,'b','LineWidth',1.5)
hold on
plot(mobileX,d2,'r','LineWidth',1.5)
xline(handoffX,'--k','Handoff')
grid on
xlabel('Mobile position X (km)')
ylabel('Distance to cell center (km)')
title('Mobile Handoff Simulation')
legend('Cell 1 distance','Cell 2 distance','Location','best')
hold off
```

Implementation:

Command Window

Radius_km	CellArea_km2	NumberOfCells	TotalUsers
1	2.5981	2	60
0.5	0.64952	8	240

>>



Experiment 3 :

```
clear

clc


freqGHz = [0.7 3.5 28];

freqMHz = freqGHz*1000;

bandName = ["Sub-1 GHz"; "Mid-band"; "mmWave"];


txPower = 30;

gt = 0;

gr = 0;


distance = logspace(log10(0.1),log10(2),200);

d = 0.5;


fspl = @(f,x) 32.44 + 20*log10(f) + 20*log10(x);
```

```
loss = fspl(freqMHz,d);
rxPower = txPower + gt + gr - loss;

Result = table(bandName,freqGHz',loss',rxPower', ...
    'VariableNames',{'Band','Frequency_GHz','PathLoss_dB','RxPower_dBm'});
disp(Result)

set(groot,'DefaultFigureVisible','on');

figure
hold on
grid on
for k = 1:length(freqGHz)
    plot(distance,fspl(freqMHz(k),distance),'LineWidth',1.5)
end
set(gca,'XScale','log')
xlabel('Distance (km)')
ylabel('Path Loss (dB)')
title('Free-Space Path Loss')
legend(bandName,'Location','northwest')
hold off

figure
hold on
grid on
for k = 1:length(freqGHz)
    rx = txPower + gt + gr - fspl(freqMHz(k),distance);
    plot(distance,rx,'LineWidth',1.5)
end
set(gca,'XScale','log')
xlabel('Distance (km)')
ylabel('Received Power (dBm)')
title('Received Power vs Distance')
legend(bandName,'Location','southwest')
```

```
hold off

N = [1 3 7 12];
n = 3.5;
sir = (sqrt(3*N).^n)/6;
sirDb = 10*log10(sir);

figure
plot(N,sirDb,'o-','LineWidth',1.5)
grid on
xlabel('Reuse Cluster Size (N)')
ylabel('Estimated SIR (dB)')
title('Frequency Reuse vs Cellular Interference')

disp(table(N',sirDb', ...
    'VariableNames',{'ReuseClusterSize','SIR_dB'}))
```

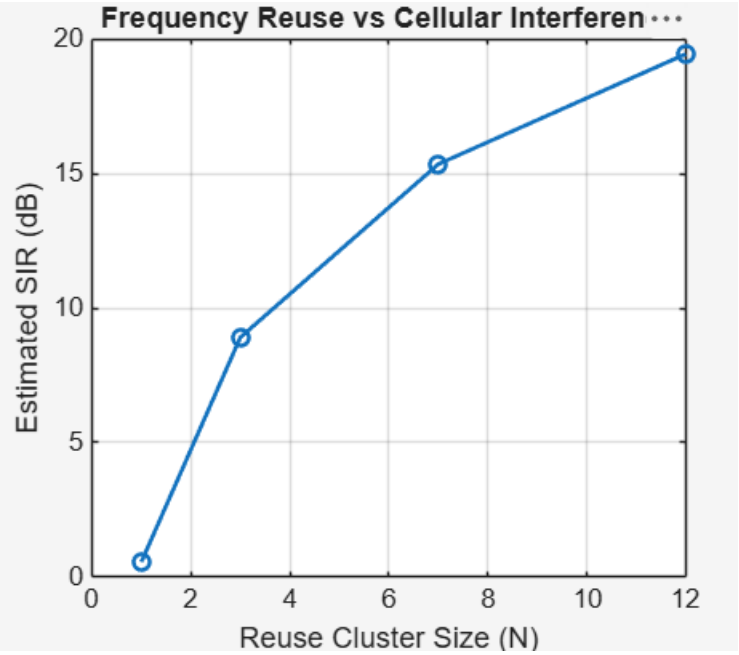
Implementation:

Command Window

Band	Frequency_GHz	PathLoss_dB	RxPower_dBm
"Sub-1 GHz"	0.7	83.321	-53.321
"Mid-band"	3.5	97.301	-67.301
"mmWave"	28	115.36	-85.363

ReuseClusterSize	SIR_dB
1	0.56811
3	8.9177
7	15.357
12	19.454

>>



Experiment 4:

```
clc
clear

distance = input('Enter 5 distances in meters: ');
RSSI = input('Enter 5 measured RSSI values in dBm: ');
APcount = input('Enter visible AP counts at 5 locations: ');

figure
plot(distance,RSSI,'o-','LineWidth',1.5)
grid on
xlabel('Distance (m)')
ylabel('RSSI (dBm)')
title('Wi-Fi Signal Strength vs Distance')

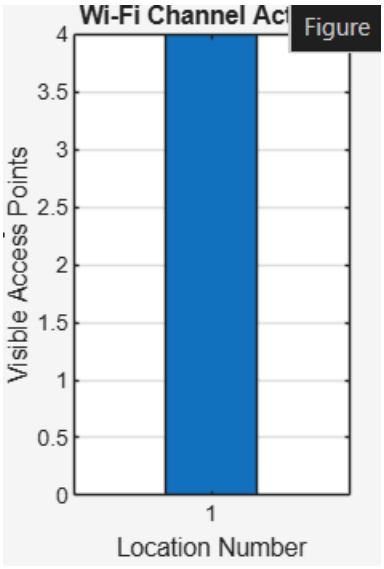
figure
bar(APcount)
grid on
xlabel('Location Number')
ylabel('Visible Access Points')
title('Wi-Fi Channel Activity')
```

```
disp('Distance and RSSI measurements:')
disp(table(distance(:),RSSI(:), ...
    'VariableNames',{'Distance_m','RSSI_dBm'}))
```

Implementation:

```
Command Window
Enter 5 distances in meters: 4
Enter 5 measured RSSI values in dBm: 23
Enter visible AP counts at 5 locations: 4
Distance and RSSI measurements:
    Distance_m    RSSI_dBm
    __________    ________

        4             23
```



Experiment 5:

```
clc;
clear;
close all;

BW1 = 20;
SE = 4;
Ncc = 1;
```

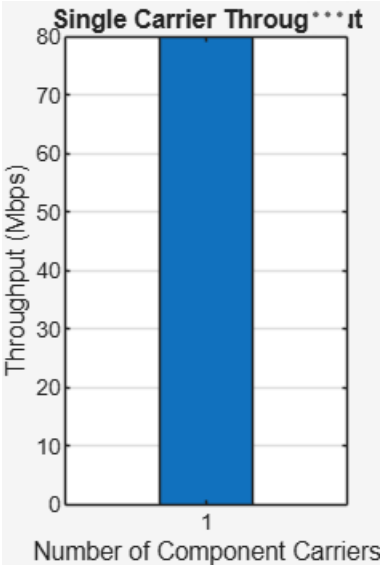
```
BWagg = BW1*Ncc;
TP = BWagg*SE;

disp(['Bandwidth = ',num2str(BWagg),' MHz'])
disp(['Throughput = ',num2str(TP),' Mbps'])

figure;
bar(Ncc,TP);
grid on;
xlabel('Number of Component Carriers');
ylabel('Throughput (Mbps)');
title('Single Carrier Throughput');
```

Implementation:

```
Command Window
Bandwidth = 20 MHz
Throughput = 80 Mbps
>>
```



Experiment 6:

```
clc;
clear;
close all;
```

```
SNR = 0:2:20;
BER1 = zeros(size(SNR));
BER2 = zeros(size(SNR));

for k = 1:length(SNR)
    N = 10000;
    bits = randi([0 1],N,1);
    tx = 2*bits-1;

    snrLin = 10^(SNR(k)/10);
    noise = randn(N,1)/sqrt(2*snrLin);
    rx = tx+noise;

    bhat = real(rx)>0;
    BER1(k) = mean(bhat~=bits);

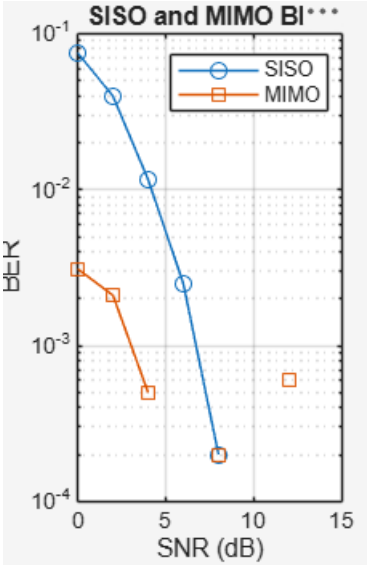
    H = (randn(2,2)+1i*randn(2,2))/sqrt(2);
    h = sqrt(sum(abs(H(:)).^2)/2);
    noise2 = randn(N,1)/sqrt(2*snrLin);
    rx2 = h*tx+noise2;

    bhat2 = real(rx2)>0;
    BER2(k) = mean(bhat2~=bits);
end

figure;
semilogy(SNR,BER1,'o-');
hold on;
semilogy(SNR,BER2,'s-');
grid on;
xlabel('SNR (dB)');
ylabel('BER');
```

```
title('SISO and MIMO BER');
legend('SISO','MIMO');
```

Implementation:




```
Command Window
>> 9

ans =

     9

>>
```

Experiment 7:

```
clc;
clear;
close all;


Nsc = 24;
Nsym = 14;


G = zeros(Nsym,Nsc);
G(1:10,:) = 1;
G(11:14,:) = 2;
```

```
figure;
imagesc(G);
grid on;
xlabel('Subcarrier Index');
ylabel('OFDM Symbol Index');
title('5G NR
Resource Grid');
colorbar;
```

Implementation:

